

Question 1

Prove that $x^* = (1, 1/2, -1)$ is optimal for the optimization problem,

$$\begin{aligned} &\text{minimize } (1/2)x^T P x + q^T x + r \\ &\text{subject to } -1 \leq x_i \leq 1, \quad i = 1, 2, 3 \end{aligned}$$

where,

$$P = \begin{bmatrix} 13 & 12 & -2 \\ 12 & 17 & 6 \\ -2 & 6 & 12 \end{bmatrix}, \quad q = \begin{bmatrix} -22.0 \\ -14.5 \\ 13.0 \end{bmatrix}, \quad r = 1.$$

$$f(x) = \frac{1}{2} x^T P x + q^T x + r$$

$$\nabla f(x) = P x + q$$

According to Optimality Criterion:

$$\nabla f(x^*)^T (y - x^*) \geq 0 \quad \text{for all feasible } y$$

$$\Leftrightarrow \text{if } x^* \text{ is optimal, satisfy } \nabla f(x^*)^T (y - x^*) \geq 0$$

$$\text{Need Prove: } \Rightarrow (P x^* + q)^T (y - x^*) \geq 0 \quad (1)$$

$$\left\{ \begin{bmatrix} 13 & 12 & -2 \\ 12 & 17 & 6 \\ -2 & 6 & 12 \end{bmatrix} \begin{bmatrix} 1 \\ 0.5 \\ -1 \end{bmatrix} + \begin{bmatrix} -22 \\ -14.5 \\ 13 \end{bmatrix} \right\} \cdot \begin{pmatrix} y_1 - 1 \\ y_2 - 0.5 \\ y_3 + 1 \end{pmatrix} \geq 0$$

$$\begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}^T \begin{bmatrix} y_1 - 1 \\ y_2 - 0.5 \\ y_3 + 1 \end{bmatrix} \geq 0$$

$$\text{Prove: } (1 - y_1) + 2(y_3 + 1) \geq 0 \quad (2)$$

$$\text{each } -1 \leq y_i \leq 1$$

$$\text{so } \begin{cases} 0 \leq 1 - y_1 \leq 2 \\ 0 \leq y_3 + 1 \leq 2 \end{cases}$$

$\Rightarrow$ Obviously (2) establish

So do (1), x^* is the optimal value



Question 2

Give an explicit solution of each of the following LPs.

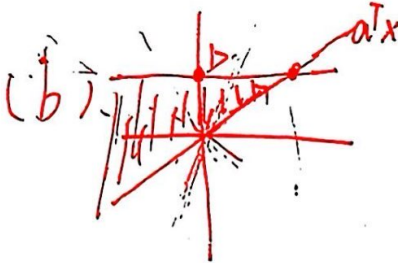
(a) Minimizing a linear function over an affine set.

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax = b \end{array}$$

hint: Suppose

(b) Minimizing a linear function over a halfspace.

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & a^T x \leq b \end{array}$$



$$L(x, \lambda) = c^T x + \lambda(a^T x - b)$$

$$\begin{aligned} \nabla_x L &= c + \lambda a = 0 \\ \nabla_\lambda L &= a^T x - b = 0 \end{aligned} \Rightarrow \begin{cases} x^* = \frac{b}{a^T a} a \\ \lambda = -\frac{c}{a} \end{cases}$$

